Pratik Relekar

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LinkedIn GitHub Portfolio

Research Interests

My research interests center on how computational methods and AI systems can help us understand human decisions and economic activity. I am especially interested in machine learning, database systems, computational social science, and econometrics.

Education

University of Illinois at Urbana–Champaign (UIUC) **Aug 2022 – May 2023**
M.S. in Business Analytics (MSBA) — GPA 3.85/4.0 *Urbana, IL, USA*

University of Pune **Aug 2010 – May 2014**
B.E. in Electrical Engineering — First Class with Distinction (GPA 3.5/4.0) *Pune, India*

Relevant Coursework
Big Data Infrastructures, Big Data Analytics in Finance, Supply Chain Analytics (machine learning), Enterprise Database Management, Financial Data Management & Analysis, Business Intelligence, and Data Storytelling.

Research Experience

Data Science Research Services (DSRS), UIUC **Jul 2026 – Present**
Research Fellow | PI: Prof. Matias Carrasco Kind *Urbana, IL, USA*
Consumption–Nightlights Settlement Detection (VIIRS × NielsenIQ)

- Investigating whether product-level consumption can distinguish genuine, permanent new settlement from transient activity that nighttime-lights imagery alone cannot separate.
- Built a balanced 205-DMA panel of NielsenIQ retail-scanner consumption and ran population- and income-controlled long-difference regressions. Standard covariates explain roughly 5–15% of cross-DMA consumption variation, leaving residual variation to compare with VIIRS spatial-structure measures.
- Conducting topological analysis of VIIRS nighttime-lights imagery (Betti curves) and assessing how spatial structure relates to the consumption residual.

Data Science Research Services (DSRS), UIUC **May 2023 – Dec 2024**
Data Scientist (Research) *Urbana, IL, USA*

- Designed and implemented a multimodal Document AI pipeline leveraging Microsoft LayoutLMv3 to extract dense financial tables and disclosures from 2,000+ image-based PDFs (tax documents and 10-K filings), achieving full spatial and semantic preservation of structural accountancy data.
- Built a financial NLP pipeline using FinBERT to extract tone and adapted BloombergGPT architectures to model sentiment in management disclosures. Reached 83% accuracy in linking disclosures to stock-market reactions.
- Built and analyzed a 12-year (2011–2022) longitudinal panel of Illinois school- and district-level records (EdFacts and CCD via the Urban Institute, and ISBE State Report Card), harmonizing heterogeneous formats into analysis-ready datasets.
- Conducted a causal-inference study estimating how educational attainment affects cannabis consumption, linking the education panel to county-level cannabis-use and socioeconomic data across Illinois households.

Publications & Open-Source Software

NielsenIQRetail (NielsenDSRS) — *Python* *Oct 2026 (v1.0.0 release)*

- Built an open-source Python library for parallel, out-of-core reading of NielsenIQ Retail Scanner data. Its Dask-based interface reads both data eras (2006–2020 and 2021 onward).
- Available on PyPI and GitHub.

Professional Experience

Johnson Controls Inc. **Jan 2025 – Present**
Data Scientist *Chicago, IL, USA / India*

- Built a Customer-Master entity-resolution system reconciling 1.7M+ messy customer records into organizational hierarchies (branches, dealers, and subsidiaries rolling up to parent entities). Matched records using Snowflake Arctic and Voyage multilingual-2 embeddings and a field-weighted similarity score tuned through supervised optimization. Reached 90%+ accuracy and identified the 20 accounts that drive 80% of Global Products revenue.
- Designed and benchmarked a multi-stage address-resolution pipeline over 1M+ customer addresses, selecting Microsoft Azure Maps after comparing accuracy and cost against probabilistic and transformer-embedding baselines. The pipeline used Snowflake regex preprocessing, Azure Maps geocoding, and a second cleanup pass to reach a 94.6% clean-data rate. Routed low-confidence records into an active-learning review loop for geographic market-share analysis.
- Built a retrieval-augmented generation (RAG) agent in Microsoft Prometheus to monitor weekly infrastructure news on 20 prospective data-center customers. Tracked investment, footprint, and power capacity to identify early bid opportunities. Semantic filters on Azure GPT models reduced irrelevant context by more than 90%.

Johnson Controls International PLC **Nov 2021 – Jul 2022**
Business Analyst *India*

- Applied named entity recognition with fuzzy matching to deduplicate and cleanse 1M+ CRM records, improving sales-lead identification by 30%.
- Built SQL ETL pipelines (stored procedures) to normalize data to regional and departmental standards, improving query speed by 40%.
- Optimized SQL powering Power BI sales dashboards, reducing refresh times by 20%.

Indium Software Pvt. Ltd. **Jun 2021 – Oct 2021**
Data / Business Intelligence Analyst *India*

- Designed 50+ Tableau dashboards delivering customer insights, lifting client engagement and conversion rates by 15%.

Skymet Weather Services Ltd. (Dept. of Agriculture, Maharashtra) **Nov 2019 – May 2021**
Data Analyst *India*

- Trained regression models (random forest, gradient boosting, and neural networks) on rainfall, wind, humidity, and soil parameters across 25+ districts to generate crop-advisory predictions.
- Developed a two-stage loan-default forecasting system. A gradient-boosted model projected each farmer's next-season borrowing from crop advisories and prior borrowing, and a classifier estimated default probability using those projections and repayment history.
- Built a soil-health classification model using ensemble methods at 85% accuracy and migrated agricultural data from flat files to PostgreSQL, making retrieval 75% faster.

Bharat Forge Ltd. & Trimurti Stampings Pvt. Ltd. **Jan 2015 – Oct 2019**
Design / Electrical Engineer — automotive electrical systems for vehicle platforms *India*

Teaching & Mentorship

AFOSOP Institute **May 2017 – Oct 2019**
Visiting Mentor — Armed Forces aspirants *India*

- Mentored candidates preparing for Armed Forces selection, covering the written examination, quantitative aptitude, and the Services Selection Board (SSB) interview.

Honors

- Top 10% of MSBA cohort, University of Illinois at Urbana–Champaign.
- All India Rank 5, Armed Forces engineers' entry examination (2016).

Technical Skills

Programming	Python (pandas, Polars, NumPy, scikit-learn, PyTorch, Dask), SQL
ML & NLP	Large language models (LLMs), retrieval-augmented generation (RAG), text classification, OCR pipelines, spaCy, NLTK
Data & Systems	Snowflake, PySpark, PostgreSQL, MySQL, AWS, ETL
Spatial & TDA	VIIRS nighttime lights, geospatial panels, topological data analysis (Betti curves)
Visualization	Tableau, Power BI, Matplotlib, Seaborn, ggplot